

Pulkit Garg

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EDUCATION

University of Waterloo - Bachelor of Computer Science Sep 2022 - Present
Cognitive Science Minor · Fine Arts Specialization · President's Scholarship of Distinction

TECHNICAL SKILLS

Languages: C++, C#, Python, Java, Kotlin, GLSL, HLSL

Graphics & Systems: Vulkan, OpenGL, GPU Programming, Compute Shaders, Multithreading, Real-Time Rendering

Tools & Frameworks: Git, CMake, Android SDK, Unity, Unreal Engine, React Native

EXPERIENCE

Graphics Programming Intern - SideFX Jan 2026 - Present

- Implemented core 3D viewport functionality in Vulkan and built multiple Vulkan libraries with custom GLSL shaders, achieving more than 20x polygon reduction across instances and significant draw-call reduction.
- Designed and built a GPU-based Vulkan text-rendering pipeline from scratch using bitmap fonts and texture atlases; implemented skyline bin-packing and extended the system to support TrueType font rasterization.
- Worked directly on performance-critical, cross-platform rendering code used by professional VFX artists in Houdini.

3D Software Development R&D Intern - SideFX May 2025 - Sep 2025

- Improved data import systems by enabling high-precision numerical parsing and processing for scientific and simulation workflows.
- Reworked mesh decimation algorithms to achieve more than 10x polygon reduction while preserving topological detail and UV/texture maps.
- Designed and implemented a real-time geometry editing tool based on the Push-Pull algorithm, using custom spatial acceleration structures to reduce user interaction complexity by up to 12x and increase modeling throughput by 30%.

Virtual Reality Developer - Liftwerx Sep 2024 - Dec 2024

- Developed VR training modules for wind turbine technicians using Unity, Unreal Engine, and Blender.
- Led end-to-end development including scripting, debugging, and deployment on AWS S3 and Meta Quest headsets using a custom Android launcher.
- Created a custom Android plugin enabling centralized data synchronization and real-time reporting on technician performance.

Software Developer Intern - OS & Graphics - Ford Motor Company of Canada Jan 2024 - Apr 2024

- Contributed to Android Automotive OS by integrating critical vehicle systems including lighting and climate using C++ and Kotlin.
- Reduced memory leaks by refactoring static companion objects, improving OS stability.
- Reduced service load times by up to 70% by implementing multithreaded services.

Software Developer Intern - 3D Visualization - Ford Motor Company of Canada [May 2023 - Sep 2023]

- Built real-time 3D visualization tools for next-generation Ford vehicles using Google Filament; implemented asynchronous asset loading to improve load times, graphical fidelity, and user experience.

PROJECTS

FluidSim - C++, Vulkan, GLSL GitHub

Highly optimized real-time GPU particle-based fluid simulation with photorealistic rendering, running on a custom-built Vulkan graphics engine.

Voxel - C#, Unity, HLSL GitHub

Cross-platform open-source 3D modeling tool for voxel-based game asset creation.

Pashu - Unity, C#

Real-time strategy game about running an animal shelter, featuring animal AI, resource management, and a time-of-day system; featured in the GitHub Education program portfolio.

SpaceLine - Unity, SteamVR

Physics-based space sandbox with real-time gravity and climate simulation, plus a VR mode built with SteamVR; won NASA Space Apps India and was selected as a global finalist.

AWARDS

Hawk Hacks 2023 - Laurier University · Hack Western 2022 - Best Gaming Hack

NASA SpaceApps 2019 - National Winner & Global Finalist among 30,000 participants · AngelHack 2019 -

AWS Award